```
Andrii Dzhus, IT-41, N = 10
--- Task 1 ---
[ 1  2  3  4  5  6  7  8  9 10]
shape: (10,)
size:  10
dtype: intó4
--- Task 2 ---
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
[[1 1 1 1 1 1 1 1 1 1]
 [1 1 1 1 1 1 1 1 1 1]]
[[10 10]
 [10 10]]
[[1. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 1. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 1. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 1. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 1. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 1. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 1. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 1. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 1.]]
--- Task 3 ---
[0.         0.11111111 0.22222222 0.33333333 0.44444444 0.55555556
 0.66666667 0.77777778 0.88888889 1.        ]
[ 0  2  4  6  8 10 12 14 16 18 20]
```

```
--- Task 3 ---
[0.         0.11111111 0.22222222 0.33333333 0.44444444 0.55555556
 0.66666667 0.77777778 0.88888889 1.        ]
[ 0  2  4  6  8 10 12 14 16 18 20]
--- Task 4 ---
[ 1  2  3  4  5  6  7  8  9 10 11 12]
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
(3, 4)
[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [10 11 12]]
(4, 3)
[[ 1  2  3  4  5  6]
 [ 7  8  9 10 11 12]]
(2, 6)
--- Task 5 ---
[ 2  4  6  8 10 12 14 16 18 20]
[11 12 13 14 15 16 17 18 19 20]
[  1   4   9  16  25  36  49  64  81 100]
[0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
385
```

```
--- Task 6 ---
6.5
14.5
10.714285714285714
6
[False  True  True False False  True  True]
4
[47.3 51.8 56.3 48.2 43.7 53.6 58.1]
--- Task 7 ---
78
[15 18 21 24]
[10 26 42]
```